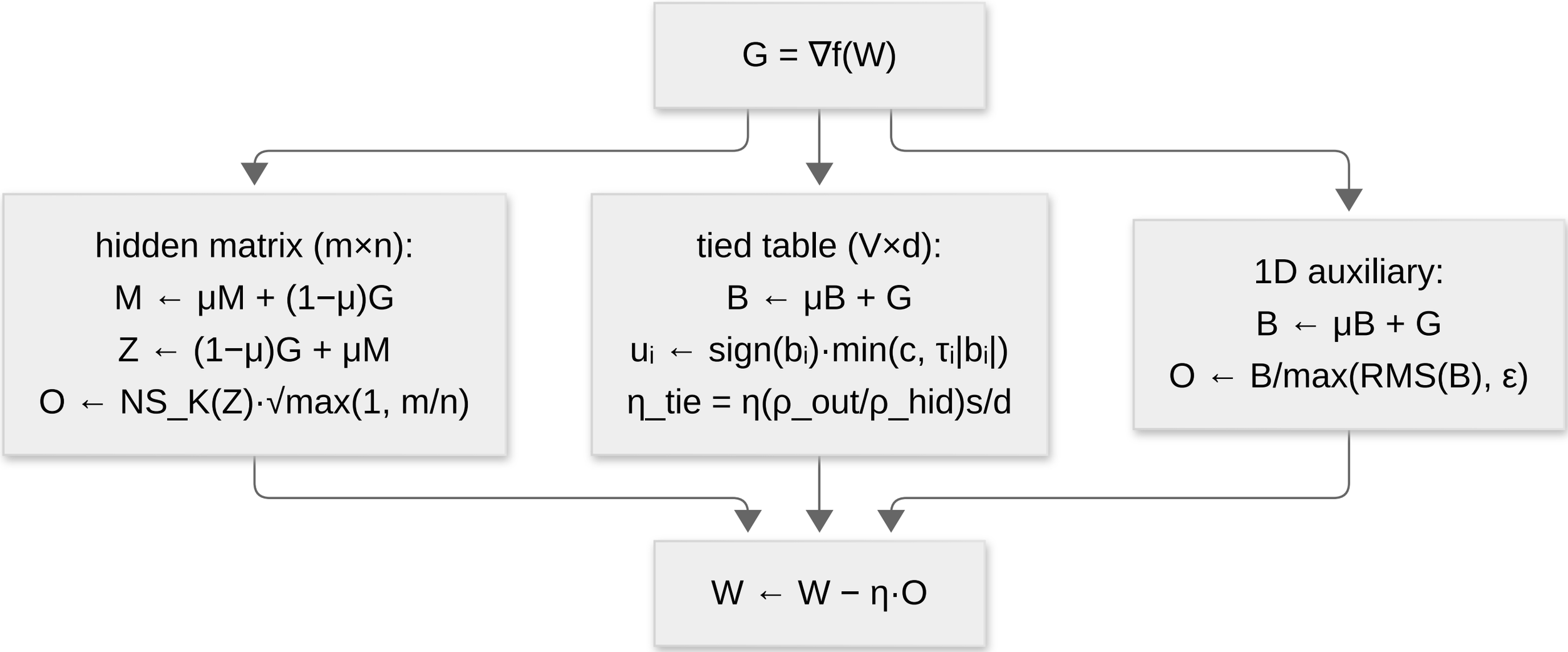

G = ∇f(W)
hidden matrix (m×n):
M ← μM + (1−μ)G
Z ← (1−μ)G + μM
O ← NS_K(Z)·√max(1, m/n)
tied table (V×d):
B ← μB + G
u_i ← sign(b_i)·min(c, τ_i|b_i|)
η_tie = η(ρ_out/ρ_hid)s/d
1D auxiliary:
B ← μB + G
O ← B/max(RMS(B), ε)
W ← W − η·O